# Stage 4 — Hyperbolic window: an indefinite block (a ghost)

Fake Weil polynomial x² − 5x + 5 at q = 5 (a² = 25 > 20; TAA Lemma 152). Colour = sign(deg)·√|deg|: orange deg > 0, purple deg < 0 (impossible for a real curve).

nested rings = weight strata |z| = q^(−σ)
= attenuation classes; purity = one ring
= distortionless (÷ q^½ → lossless)
[Restatement for Frobenius / Analogy for ζ]

y-axis = critical line Re s = ½:
the unitary spectrum of scaling
U_λ f(x) = λ^½ f(λx) on half-densities
(|λ^(½−s)| = 1 ∀λ ⇔ Re s = ½; TAA Lemma 153 §5)
[Theorem; frame axis = Analogy]

null rays = asymptotes m = r·n, r = (a ± √(a² − 4q))/2
= 3.618, 1.382: F and V are two different REAL numbers;
real absolute ⇒ boosts, Krein-neutral pair = ghost
[Theorem for fakes]

LC with C < 0: energy hyperbola (runaway mode).
Wick rotation t → iτ: dx² − c²dt² → dx² + c²dτ²
(OS link; a different sign flip from C → −C, TAA Lemma 153, correction 30)
[Theorem for the conic; Analogy for the link]

hyperbolic window = indefinite 2×2 block.
DH prime-side tilt blocks (w = 1.5) are hyperbolic on
[84.36, 84.77], [85.48, 85.91], [114.04, 114.29],
near the off-line heights 85.699, 114.163; ζ: none on [10,120]
(TAA Lemma 153, significant from X = e^12)
[Numerical]

ghost = split resonance doublet: DH ghost 1 (β = 0.8085) cusp HWHM 0.0957 / 0.4042
(predicted (1−β)/2, β/2 = 0.0957 / 0.4043; TAA Lemma 154);
seen from σ = ½ + d < β the line inverts (gain line)
[Numerical]

origin ↔ point at infinity = the cusp:
constant term $a_0(y) = y^s + \varphi(s)\cdot y^{(1-s)}$,
|φ(½+it)| = 1 (TAA Lemma 153 NUM ≤ 2e−31)
[Analogy for the identification]

Legendre support window < π per branch:
2·atan√(|λ₋|/λ₊); empty for the full level set
[Theorem]

Status tags: [Theorem] [Restatement] [Analogy] [Numerical] [Cited] — see legend.png. RH is not proved. No null rays at 45° (per J. W. McGreevy); lobes are elliptical conic sections, not elliptic curves.

After J. W. McGreevy's animation 'Flat Potential Oscillations' (Aug 2025, x.com/McGreevyMD/status/1951474097990279331). Re-rendered by Grok Bot from the analysed geometry; not J. W. McGreevy's code.